The Ladies' Diary

Beyond A-level

Fifty-five problems, 1709–1816,
in curves, numbers, the Earth and motion

A charming brisk maid has assur'd me and said,
Since I'm such a fine mathematician,
She will wed me—but on this condition.

John Landen, question 279

Working draft

A word before the problems

By the 1740s the Diary's hardest questions had stopped being puzzles. John Landen, a land agent from Peterborough, set a question in verse about a lady's fortune and answered other people's with the calculus of fluxions. William Emerson signed himself *Merones*. Thomas Simpson, a weaver's son who became mathematical master at the Royal Military Academy, Woolwich, and edited the Diary from 1754 to 1760, signed himself *Kubernetes*, *Hurlothrumbo* and *Patrick O'Cavanah*.[1] Sixty years later the same pages carried Peter Barlow and Olinthus Gregory on the line of quickest descent and the shape of the Earth. The Diary asked for no degree. It printed the best answer.

The other books in this series are for school pupils, and they leave out anything beyond A-level. This one collects what they left out. It keeps the mechanics: pendulums, rolling balls, hanging chains, falling bodies and water running out of a vessel. Astronomy still stays out, because it needs the tables of the day. Where the Diary asked the same thing twice, one version is here. Where a question depends on a figure that cannot be rebuilt with confidence from the words, it is not here.

How to read the labels

Leybourn numbered the questions straight through his four volumes. Numbers up to 294 come from volume I (1704–1748), 295 to 699 from volume II (1749–1776), 700 to 1088 from volume III (1777–1802), and 1089 onwards from volume IV (1802–1816). So *Q276* and *Q1162* need no volume. Prize questions carry the year Leybourn's index gives them. *After Q1128* means the numbers or units have been changed; the answer given is then for the retelling. Chains, acres and the like are replaced by metres and hectares. Feet and inches stay where the Diary's value of gravity is in feet, and a few coins stay because the problem is about them.

Answers are at the back, with the method the Diary used and, where it helps, a modern one. Two grey boxes at the end list what the Diary got wrong, and what it simply did differently.

How this edition was made

This edition was prepared with the assistance of Claude, an AI model made by Anthropic. Under the editor's direction, Claude drafted the retellings, solutions, notes and introduction, recomputed every answer, ran optical character recognition on the source scans (using Tesseract), and typeset the text. Selection, editorial decisions and final responsibility for the text are the editor's.

[1] The identifications are Leybourn's own, in the index of names at the end of vol. IV.

Curves and solids

1. The dictionary's conoid *after Q169, by William Grimmett*

A dictionary of the day stated that a hyperbolic conoid, the solid made by turning a hyperbola about its transverse axis, bears a fixed ratio to the cylinder that encloses it. Take the transverse axis to be a and cut the conoid at a distance x from its vertex. Find the true ratio. Is it fixed?

2. The road round the hill *after Q1128, by John Bennett*

A hill is a cone. A road leaves its foot and winds exactly once round it, climbing steadily, 1 m up for every 6 m along. It is 1,600 m long and ends at a summer-house exactly half way up the hill. Find the radius of the hill's base.

3. The pond traced by a point *after the prize question for 1804, by Isaac Rowbottom*

P is the centre of a circle of radius a, and AB a fixed diameter. From P draw any line to the circle at Q. From Q drop QR at right angles to AB, and from R drop RS at right angles to PQ. As Q goes round, S traces the edge of a pond. The pond covers 12 hectares. Find a, and the greatest distance of the pond's edge from AB.

4. The father's curved fence *after Q1157, by W. Z.*

A father leaves a round field to his son and daughter. The fence between them runs from one end A of a diameter AB to the other. At a distance x from A along AB, the fence stands off the diameter by the length of the chord from A to the circle at that point, less x. The son's share is worth £30,000 more than the daughter's, and the land is worth £5,000 a hectare. How large is the field?

5. The garden in the quadratrix *Q1219, by William Watkins*

CA and CB are radii of a quadrant, 10 m long, at right angles. The radius CB turns steadily about C down to CA. In the same time a line parallel to CA slides steadily from B down to CA. Their crossing point traces the quadratrix of Dinostratus. A rectangular garden has one corner at C, two sides along CA and CB, and the opposite corner on the quadratrix. Make it as large as possible.

6. Chord less abscissa *Q1248, by J. A. Dotchen*

AB is the diameter of a semicircle, of length a. D is any point of AB, and the perpendicular at D meets the circle at E. On DE take DP equal to the chord AE less AD. Find the length of the curve traced by P from A to B, without series.

7. The trapezoid in the hyperbola *after Q1271, by A. Nesbit*

A hyperbola has transverse axis 120 m. At 40 m from its vertex, a double ordinate is 96 m long. Inscribe in the hyperbola the largest trapezoid that has this double ordinate as one of its parallel sides and another double ordinate as the other.

8. The path on the turning triangle *Q1296, by George Ducket*

A right-angled triangle ABC has hypotenuse $AC = 6$ ft and $BC = 3$ ft at right angles to the base AB. It turns once about AB, steadily, while a body moves steadily along AC from A to C in the same time. How long is the body's path?

9. Sisyphus's stone *prize question for 1747, by Samuel Bamfield*

Sisyphus's stone is a cylinder 2 ft across. He rolls it straight up a hemispherical mountain half a mile high, and straight down again. How far does a spot on the rim of the stone travel?

10. Tangents to an ellipse *Q1142, by Olinthus Gregory*

A polygon $ABCDEF$ is drawn round an ellipse, its sides touching the ellipse at a, b, c, d, e, f. Show that the products of alternate segments of the sides are equal:

$$Aa \cdot Bb \cdot Cc \cdot Dd \cdot Ee \cdot Ff = aB \cdot bC \cdot cD \cdot dE \cdot eF \cdot fA.$$

Numbers

11. Counting the solutions

Q267, by John Powle

In how many ways can $35x + 43y + 55z = 4000$ be solved in positive whole numbers?

12. The charming brisk maid

Q279, by John Landen

Leybourn prints it as the Diary did:

A charming brisk maid has assur'd me and said,
Since I'm such a fine mathematician,
(Laying puzzling aside, for the joys of a bride)
She will wed me—but on this condition:
That I first shall unfold what pieces of gold
Her father has for her in store:
And these I must find from the data subjoin'd,
And then I'm to puzzle no more.

The pieces are half guineas (10s 6d), guineas (21s), moidores (27s) and three-pound-twelves (72s), 4,000 in all. If there are v, x, y and z of each, $v^4x^3y^2z$ is as large as it can be. What is the lady's fortune?

The Earth

13. If the Moon stood still

Q1094, by T. Swanwick

If the Moon lost its motion and fell straight to the Earth, how long would it take? Take the Earth's radius as 3,965 miles, the Moon's distance as 60 radii, and gravity at the surface as $32\frac{1}{6}$ feet per second per second.

14. The plumb line and the ground

Q1154, by William Wiseman

Take the Earth to be an oblate spheroid with axes in the ratio 230 to 229. A plumb line points to the centre; the ground is level at right angles to the normal. In what latitude is the angle between the line to the centre and the normal greatest, and how large is it?

15. One degree against two

Q1220, by William Wright

On the same spheroid, in what latitude is a degree of the meridian as long as two degrees of longitude?

16. The density of the Earth

Q1279, by James North

Charles Hutton had found that the Earth's mean density is about $\frac{9}{5}$ of the density of the rocks at its surface. Suppose the density at distance r from the centre varies as r^n. Find n.

Motion and force

17. A pendulum that counts itself

Q10

Find the length of a pendulum that makes as many swings in a minute as it is inches long. The seconds pendulum is 39.2 inches.

18. The clockmaker's pendulum

Q1110, by William Francis, junior

A clockmaker fits a new pendulum to an old clock, which then gains 5 minutes in 12 hours. He lengthens the pendulum by 2 inches, and now it loses 5 seconds an hour. What is the right length, and how many swings does it make a minute? Take the seconds pendulum as $39\frac{1}{8}$ inches.

19. The conical pendulum

Q276, by John Landen

A pendulum swings round in a cone whose height is 200 inches. Find the time of one revolution.

20. Three elastic balls

Q243, by John Powle

Three perfectly elastic balls A, B, C lie in a line. A weighs 3 lb and C 27 lb. A strikes B, which then strikes C. What must B weigh for C to move as fast as possible?

21. The hanging chain

Q244, by John Turner

A chain 18 inches long hangs from two pins 12 inches apart on the same level. How far below the pins is its lowest point, and what area lies between the chain and the line of the pins?

22. The chain on the table

Q1282, by W. G. Horner

A chain 67 inches long hangs from a point 40 inches above a table, the rest lying straight on the table. A loose piece of the same chain only starts to slide when the table is tilted to $16°40'$. How much chain lies on the table when it is in equilibrium, with as little as possible hanging?

23. Where to strike

Q284, by Cuthbert Cockson

A man swings a uniform stick 50 inches long, his arm 20 inches long, turning about the shoulder. With what part of the stick should he strike to give the greatest blow?

24. The rolling ball

Q1172, by the Rev. J. Furnass

A lead ball is dragged up a plane inclined at 30° to the top, 50 yards high. The chain breaks and the ball rolls down without slipping. How long does it take, and how fast is it going at the bottom?

25. The trundled wheel

Q1143, by Peter Barlow

A wheel 5 feet across is rolled half a turn along level ground and back again, four times a minute, for a year of 365 days. At the start of each roll two points of its rim are level with the centre. How much farther does one of them travel than the other in the year?

26. Two balls, two planes

Q1293, by William Burdon

Two straight planes meet at a point, inclined to each other at 15°. Two balls are let go together from that point, one down each. When one is exactly above the other they are 50 feet apart. Find the inclination of each plane to the horizon, and the time.

27. Quickest and longest descent *Q1102 and Q1144, by J. G.*

(a) Two circles lie in a vertical plane, one wholly above the other. Find the straight line down which a body slides from the upper circle to the lower in the least time. (b) A point lies above a circle. Find the straight line from the point to the circle down which a body takes the longest time.

28. Over the top *Q1222, by William Moore*

A body is sent up an inclined plane with a given speed, and flies off the top. The base of the plane is given. At what inclination does it land farthest beyond the foot of the plane? Take the base as 3 and the height due to the speed as 10.

29. Drawn up the quadrant *prize question for 1810, by R. B.*

AB and AC are equal lines 10 feet long, AB horizontal and AC vertical, and BC is a quarter circle with centre A. A string passes freely over a pulley at C. A 9 lb weight hangs from one end; the other end is tied to a 1 lb weight at B, which is drawn up the arc to C. Find its speed on arriving, and the time taken.

30. The ring on the turning Earth *Q1162, by J. Skewes*

A slender rod 500 miles long stands upright in latitude 50°, turning with the Earth once in 24 hours. A heavy ring slides down it from the top. Taking gravity as constant, how long does it take? Take the Earth's radius as 3,980 miles.

31. The falling cork *Q1098, by Mrs Macdonald*

A cork ball falls from a great height through air of uniform density and settles to a steady 10 feet per second. The specific gravities of cork and air are 240 and $1\frac{2}{9}$. How large is the ball? Take the air's resistance to equal the weight of a column of air as wide as the ball and half as tall as the height from which a body must fall to gain its speed. This is Newton's law, which the Diary used.

32. The vessel that empties *Q1127, by William Burdon*

A vessel is a frustum of a cone 36 inches deep, 50 inches across the top and 40 across the bottom, full of water. It empties through a hole half an inch across in the bottom, and the issuing stream contracts to two-thirds of the hole. How long does it take? If it stood on its wide end, how wide a hole would empty it in the same time?

33. The ball against the wall *Q1297, by Omicron*

Where must a person stand, in a vertical plane at right angles to a wall, so that a perfectly elastic ball thrown against a given point of the wall, striking it with a given speed, comes back to them?

34. Admiral Vernon's chase *prize question for 1742, by Robert Heath*

Admiral Vernon, sailing due south from Jamaica towards Cartagena, sees a Spanish ship dead ahead, 7 leagues off, steering due west along the shore. He keeps his bow pointed straight at it all the way. When he catches it, the Spaniard has sailed 8 leagues. Both ships keep a steady speed. How far has Vernon sailed?

35. The spider and the fly *Q310, by J. Ash*

A spider sits at one corner of a semicircular pane of glass. A fly walks steadily along the curved edge towards the far corner, starting 30° of arc from the spider's corner. The spider runs steadily, always straight at the fly, and catches it exactly at the far corner. Find the ratio of their speeds.

36. The ball that never comes back *Q300, by Rodomontado*

With what least speed must an iron ball weighing 12 pounds be fired from the surface of the Earth, at an elevation of 40°, so that it never returns? Ignore the air, take the Earth's radius as 21 million feet and gravity at the surface as 32.2 feet per second per second.

37. Through the middle of the Earth *Q784, by Terricola*

A shaft is bored straight through the Earth along a diameter, and a ball is dropped into it from the surface. Treat the Earth as a uniform sphere 8,000 miles across, ignore its turning and the air. How long does the ball take to reach the centre, how fast is it then going, and what happens next?

38. Falling from the Sun *Q855, by X. + Y.*

A ball starts at rest 100 million miles from the Earth, the distance of the Sun, and is pulled by the Earth alone. How long does it take to reach the Earth's surface, and how fast is it going when it gets there? Take the Earth's radius as 21 million feet.

39. The report of the shell *Q484, by Richard Gibbons*

A shell is fired from a mortar at an elevation of 30°. The sound of its landing is heard at the mortar exactly 20 seconds after the firing. Find the range. Take sound to travel 1,142 feet a second, and a falling body to drop $16\frac{1}{12}$ feet in the first second. Ignore the air.

40. The pillar that unwinds *Q322, by John Landen*

A solid stone pillar, a cylinder 2 yards round, is being hoisted, and hangs 20 yards above the ground. A rope is wound ten times round its middle, with its end fixed to the stone; the other end runs up over pulleys. The fixing gives way, and the pillar falls, unwinding itself from the rope as it goes. How long does it take to reach the ground?

41. The chain over the pulley *prize question for 1768, by Thomas Allen*

A chain 40 feet long, of very small equal links, hangs over a frictionless pulley, one end 21 feet below it and the other 19. Left to itself, it starts to run. How long before the long end reaches a floor 200 feet below the pulley? Take $g = 32\frac{1}{6}$ feet per second per second.

42. Two weights on a thread *Q402, by W. Bevil*

The ends of a thread 10 feet long are tied to two tacks 6 feet apart on the same level. Where must weights of 3 and 5 ounces be fixed to the thread so that both hang at rest 3 feet below the level of the tacks?

43. The chain that pulls least *Q417, by W. Bevil*

A chain is to hang from two tacks on the same level, 10 feet apart. How long should it be so that the pull on each tack is as small as possible?

44. Pulling down the tree *Q901, by John Cullyer*

A carpenter has nearly felled a tree 60 feet high. He has a rope 70 feet long, and will pull on it standing on the ground. How high up the tree should he tie it, so that he brings the tree down with the least effort?

45. The rod against the wall *after Q433, by R. Weston*

A uniform rod leans against a smooth vertical wall, its foot on a smooth horizontal floor, and is let go from rest. Its foot slides away from the wall and its top slides down. Show that the top leaves the wall when it has fallen to two-thirds of its first height.

46. The cone that counts its swings *Q576, by William Sewell*

A solid cone holding 1,728 cubic inches hangs by its point and swings as a pendulum. It makes as many swings a minute as it is inches tall. Find its height and the radius of its base. Take the seconds pendulum as 39.2 inches.

47. The best place to hang a cylinder *Q786, by Plus Minus*

A solid cylinder swings as a pendulum about a horizontal axis through a point of its own axis. Find the length of the simple pendulum that keeps time with it. At what point should it hang to swing fastest?

48. The pull on the nail *Q841, by the Rev. John Hellins*

A lead ball hangs from a nail by a thread and swings through a given arc. What is the greatest sideways pull it puts on the nail? Treat the ball as a point.

49. The stick in the thunderstorm *Q888, by John Cullyer*

In a thunderstorm I held a uniform stick 5 feet long by one end and swung it round in a cone, its free end tracing level circles 6 feet across. It went round 7 times between the lightning and the thunder. How far away was the storm? Take sound to travel 1,142 feet a second.

50. The cracking wheel *prize question for 1797, by John Rodham*

I am walking at 4 miles an hour along a straight road. A coach overtakes me, going the same way at a steady speed. One of its wheels cracks once every turn. Before it passed me I heard 70 cracks a minute; afterwards only 68. Allowing for the speed of sound, 1,142 feet a second, how fast was the coach going?

51. Watering the wine *Q810, by the Rev. John Hellins*

A gentleman "remarkable for odd experiments" has a cask holding 100 gallons of wine. A hole is made in it, and at the same moment water is let in to keep it always full. The mixture runs out at one gallon every 10 seconds, and stays perfectly mixed. How much wine is left after an hour?

52. The cone and the sphere *Q563, by Paul Sharp*

A cone whose height equals the diameter of its base stands on its base, full of water. A hollow sphere of the same diameter is also full. Each empties through an equal small hole at its lowest point. Show that they take the same time.

53. The water clock *Q965, by John Rodham*

A vessel 5 feet deep is full of water and empties through a hole of one square inch in the bottom. The surface falls at a steady 4 inches a minute all the way down. What shape is the vessel, and how much does it hold, in ale gallons of 282 cubic inches? The Diary took the speed of the water leaving the hole to be that due to half the depth; work it on that rule, and then with Torricelli's.

54. The sponge in the cannon *Q946, by Lieut. William Dixon*

A gunner pushes a sponge, fitting the bore exactly, into a cannon 10 feet long and 5 inches across, with the touch-hole stopped. He can push with a force of 100 pounds, and the barometer stands at 30 inches. How far in can he get it? Take the air to be squeezed at a steady temperature.

55. The solid that pulls hardest *prize question for 1801, by Terricola*

A given quantity of matter is to be shaped into a solid of revolution. Find the shape that pulls hardest on a point at its surface, by Newton's law of gravitation.

Answers

Curves and solids

1. With the hyperbola $y^2 = \frac{p}{a}(ax + x^2)$, the conoid is $\pi\frac{p}{a}\left(\frac{ax^2}{2} + \frac{x^3}{3}\right)$ and the cylinder $\pi\frac{p}{a}(ax + x^2)x$. Their ratio is $\frac{a}{2} + \frac{x}{3}$ to $a + x$, or $(3a + 2x) : 6(a + x)$. It is not fixed: it runs from $\frac{1}{2}$, like the paraboloid, near the vertex, down towards $\frac{1}{3}$, like the cone, far from it.

2. About 346 m. The road meets every straight line from the summit at the same angle, so when the cone is unrolled it becomes a logarithmic spiral, halving its distance from the summit in one turn. The road rises 266.7 m, so the hill is 533.3 m high. Solving for the radius that makes the spiral 1,600 m long gives $r \approx 346.0$ m and a slant side of about 635.7 m. In the Diary, with a road of one mile, the radius was 1141.75 feet.

3. In polar coordinates about P, measuring θ from AB, $PR = a\cos\theta$ and $PS = PR\cos\theta = a\cos^2\theta$. The pond is two ovals, and its area is $\frac{1}{2}\int_0^{2\pi} a^2\cos^4\theta\, d\theta = \frac{3\pi}{8}a^2$. For 12 hectares $a \approx 319.2$ m. The half-width $a\cos^2\theta\sin\theta$ is greatest when $\tan\theta = 1/\sqrt{2}$, giving $2a/(3\sqrt{3}) \approx 122.8$ m.

4. The chord from A is $\sqrt{dx}$, so the fence is $\sqrt{dx} - x$ from the diameter. The area between them is $\int_0^d(\sqrt{dx} - x)\,dx = \frac{d^2}{6}$. The son's excess is twice this, 6 hectares, so $d^2 = 18$ hectares and the field is $\frac{\pi}{4}\cdot 18 \approx 14.1$ hectares. The answer printed in the Diary is by Mrs Elizabeth Bryhon of Helmsley.

5. About 26.16 m^2, with sides 6.03 m along CA and 4.34 m along CB. If z is the angle the moving radius makes with CA at the corner, the condition for the greatest rectangle reduces to: the length of the arc complementary to z equals $2\tan z/\sec^2 z$. This gives $z \approx 35°42'$.

6. With $AD = x$, $DP = \sqrt{ax} - x$, and the curve is an arc of a parabola. For $a = 1$ the length is

$$\tfrac{1}{8} + \tfrac{3\sqrt{5}}{8} + \tfrac{\sqrt{2}}{16}\left(\sinh^{-1}3 + \ln(1+\sqrt{2})\right) \approx 1.20216.$$

So the whole curve is $1.20216\,a$.

7. The conjugate axis is 72 m. With x measured from the centre, the other double ordinate is $2\cdot\frac{36}{60}\sqrt{x^2 - 60^2}$ and the trapezoid has area $\left(48 + \frac{3}{5}\sqrt{x^2 - 3600}\right)(100 - x)$. This is greatest at $x \approx 64.07$, giving about 2,209.1 m^2.

8. About 11.67 ft. At distance s along AC the body is $s/2$ from the axis and turns through $2\pi s/6$, so the path has element $\sqrt{1 + (\pi s/6)^2}\,ds$. Integrating from 0 to 6 gives 11.669.

9. Each turn of the stone draws one arch of an epicycloid, of length $8r(R + r)/R$ with $r = 1$ ft and $R = 2640$ ft: 8.003 feet. Going up the quarter circle of the mountain the stone turns $R/4r = 660$ times, and as often coming down. So the spot travels $1320 \times 8.003 \approx 10{,}564$ feet, just over 2 miles. This is John Landen's answer.

10. Draw semi-diameters parallel to each side. The two tangent segments from any vertex are in the ratio of the semi-diameters parallel to them. Multiply the six ratios together; every semi-diameter appears once above and once below the line, so the product of the ratios is 1.

Numbers

11. 89. The Diary's solver fixed z and solved the two-variable equation each time. Peter Barlow later gave a general count for such equations; it is printed in the appendix to Leybourn's volume IV.

12. A product $v^4x^3y^2z$ with a fixed sum is greatest when the numbers are in proportion to the powers: 1,600, 1,200, 800 and 400. That is £840, £1,260, £1,080 and £1,440, a fortune of £4,620, as the printed answer says. The powers are garbled in the source text. We have restored them as 4, 3, 2 and 1, which is the only simple choice that gives £4,620.

The Earth

13. The fall from r_0 to the surface R under an inverse-square pull gR^2/r^2 takes

$$t = \sqrt{\frac{r_0^3}{2gR^2}}\Big(\sqrt{x(1-x)} + \cos^{-1}\sqrt{x}\Big), \qquad x = R/r_0.$$

With the Diary's figures this is about 416,070 seconds, or 4 days 19 hours 34 minutes. The Diary's solvers, working from Hutton's textbook, printed 4 days 19 hours 46 minutes. We have not traced the twelve minutes.

14. For axes a and b the angle is greatest where $\tan\phi = a/b$, and there its tangent is $\frac{a^2-b^2}{2ab}$. For 230 to 229 this is about $14'58''$, in latitude $45°7\frac{1}{2}'$.

15. About $60°4'$. A degree of the meridian is proportional to the meridian's radius of curvature M, and a degree of longitude to $N\cos\phi$, the radius of the parallel. Solving $M = 2N\cos\phi$ gives a latitude a little above $60°$, the answer for a sphere.

16. The mean density of a ball whose density is $\rho_0(r/R)^n$ is $\frac{3}{n+3}\rho_0$. Setting $\frac{3}{n+3} = \frac{9}{5}$ gives $n = -\frac{4}{3}$: the density falls off inversely as the $\frac{4}{3}$ power of the distance from the centre.

Motion and force

17. The number of swings a minute is $60\sqrt{39.2/L}$. Setting this equal to L gives $L^3 = 3600\times 39.2 =$ 141,120, so $L \approx 52.063$ inches. Almost a century later the Diary set it again, as Q1089, with the seconds pendulum at $39\frac{1}{8}$ inches, and the answer came out 52.030.

18. The swings per hour were first as 3,625 to 3,600, and after lengthening as 3,595 to 3,600. Lengths go as the inverse squares of the swings, so $L_1 + 2 = L_1(3625/3595)^2$, giving $L_1 \approx 119.34$ inches. The right length is $L_1(3625/3600)^2 \approx 121.00$ inches, which swings about 34.12 times a minute. This is Peter Barlow's answer.

19. The time of one revolution is $2\pi\sqrt{h/g}$, which depends only on the height of the cone: $2\pi\sqrt{200/386} \approx 4.52$ seconds.

20. 9 lb, the mean proportional between 3 and 27. B leaves the first impact with speed $\frac{2A}{A+B}u$, and C leaves the second with $\frac{2A}{A+B}\cdot\frac{2B}{B+C}u$. This is greatest when $B^2 = AC$. John Landen answered it.

21. The chain is a catenary $y = c\cosh(x/c)$ with $2c\sinh(6/c) = 18$, so $c \approx 3.699$ inches. The lowest point is $c(\cosh(6/c) - 1) \approx 6.03$ inches below the pins. The area between chain and line is $12(c + \text{sag}) - 18c \approx 50.2$ square inches.

22. About 21.1 inches lies on the table and 45.9 inches hangs. The friction on the lying part, μ times its weight with $\mu = \tan 16°40'$, must equal the horizontal tension at the bottom of the catenary. The catenary through the support with that tension gives the equation $x = l + \mu h - \sqrt{h^2\sec^2 16°40' + 2\mu l h}$, with $l = 67$ and $h = 40$.

23. The centre of percussion of the stick about the shoulder is at $\frac{2(a^2+ab+b^2)}{3(a+b)}$ from the shoulder, with $a = 70$ and $b = 20$: 49.63 inches. That is 29.63 inches from the hand, or 20.37 inches from the far end.

24. A rolling sphere accelerates at $\frac{5}{7}g\sin\theta$ rather than $g\sin\theta$. The plane is 300 ft long, so the time is 7.23 seconds and the final speed 83.0 ft per second.

25. In half a turn one point rises over the top and down along one arch of a cycloid, while the other passes under the centre. Their paths are $10\sqrt{2}$ and $20 - 10\sqrt{2}$ ft, differing by $20(\sqrt{2}-1) \approx 8.284$ ft. Eight such motions a minute for a year gives about 34.83 million feet, or 6,597 miles.

26. A body slides down any chord from the top of a vertical circle in the same time it falls down the diameter. So the balls are always on one circle through the point, and the 50 ft between them is a chord subtending 15° at the circumference. The circle's diameter is $25/\sin 15° \approx 193.2$ ft. The planes are inclined at 37°30′ and 52°30′, and the time is that of falling 193.2 ft, about 3.47 seconds.

27. (a) Join the top of the upper circle to the bottom of the lower one. The part of this line between the two circles is the required line. (b) Join the point to the top of the circle and produce the line to meet the circle again. The whole line is the longest descent. Both rest on the fact that the chords of a vertical circle from its top point are all descended in the same time.

28. About 36°39′. Llerien set the problem again as the prize question for 1814, asking for a construction. Let B be the foot of the base AB and erect BD equal to the height due to the speed. Join AD, and on AB produced take $AS = AD$. Let M bisect DS, and let AM meet BD at C. AC is the plane. The body's parabola then has its focus at S.

29. The descending weight falls the chord $10\sqrt{2}$ ft while the other rises 10 ft, and at C both move at the same speed. So $10v^2/2 = g(9 \cdot 10\sqrt{2} - 10)$, giving $v \approx 27.5$ ft per second. The time, found by integrating numerically, is about 1.08 seconds. The Diary gave the speed and left the time as an unfinished fluxion.

30. About 405.5 seconds, or 6 minutes 45 seconds. The acceleration along the rod is $-g + \omega^2(R + s)\cos^2\lambda$, which slows the fall slightly. Without the turning the fall would take about 405.1 seconds. The Diary gave 405.9.

31. In steady fall the weight $\frac{\pi}{6}d^3(240 - 1\frac{2}{9})$ equals the resistance $\frac{\pi d^2}{4} \cdot 1\frac{2}{9} \cdot \frac{v^2}{4g}$. This gives $d \approx 0.0060$ ft, about $\frac{1}{14}$ inch or 1.8 mm. A modern drag law would give a different size. The answer is the Diary's, on the Diary's assumptions.

32. The Diary took the stream's speed to be that due to half the depth. On that assumption it empties in 1 hour 55 minutes, and inverted it needs a hole 0.538 inches across. With Torricelli's law, the speed due to the full depth, the time is 1 hour 21 minutes. The hole for the inverted vessel is the same either way, because only the ratio of times matters.

33. With a vertical wall, a ball that comes back to the thrower must strike the wall square, moving horizontally, and return along the same parabola. So the person may stand anywhere on the parabola whose vertex is the given point, with its axis vertical and its latus rectum four times the height due to the given speed.

34. About 12.23 leagues. In a chase of this kind, with the target moving at right angles to the first line of sight at distance a and the pursuer n times as fast, the target runs $b = \frac{an}{n^2-1}$. So $8(n^2 - 1) = 7n$, $n \approx 1.529$, and Vernon sails $nb \approx 12.232$ leagues. This was Merones's answer: William Emerson's.

35. About 1.160: the spider must be a little more than one and a sixth times as fast as the fly. We found this by integrating the pursuit numerically, with the fly starting 30° of arc from the spider's corner. The Diary never printed a solution. Its 1749 answer says that Mr Farrer's solution "was not true", that Landen sent a true method whose "calculus being so operose, it was not wrought out", and that the ratio was "little different from an equality". We know of no closed form.

36. About 36,800 feet a second, just under 7 miles a second, whatever the ball weighs and whatever the angle. A body moving outwards with speed v at distance r just escapes when $\frac{1}{2}v^2 = gR^2/r$, so at the surface $v = \sqrt{2gR}$. John Landen answered it this way and noted that the elevation does not matter. Nor does the weight. The setter's pen-name, Rodomontado, means a braggart.

37. Inside a uniform Earth the pull is proportional to the distance from the centre, so the ball moves in simple harmonic motion with period $2\pi\sqrt{R/g}$. With R = 4,000 miles it reaches the centre in $\frac{\pi}{2}\sqrt{R/g} \approx 21.2$ minutes, moving at $\sqrt{gR} \approx 26{,}000$ feet, or 4.9 miles, a second. It comes to rest at the far surface 42.4 minutes after it was dropped, and swings back and forth for ever.

38. It arrives at about 36,800 feet, or 6.96 miles, a second, almost exactly the escape speed of problem 36, as it must be from so far away. Its speed at distance x is given by $v^2 = 2gR^2(1/x - 1/a)$. Integrating $dt = dx/v$ from a down to R gives

$$t = \sqrt{\frac{a^3}{2gR^2}}\Big(\sqrt{\epsilon(1-\epsilon)} + \arccos\sqrt{\epsilon}\Big), \qquad \epsilon = R/a,$$

which is about 41,400 days: some 113 years. A. Whitehouse gave the velocity as 6.961 miles a second.

39. About 6,033 feet. If the range is x, the time of flight is $\sqrt{x\tan 30°/16\frac{1}{12}}$ and the sound takes $x/1142$. Their sum is 20 seconds, which gives $x \approx 6{,}033.4$ feet, with the shell 14.7 seconds in the air. Thomas Allen's printed answer is 6,033.42 feet.

40. About 2.36 seconds. The rope is wound ten times round a cylinder 2 yards round, so it holds exactly the 20 yards of the fall, and the pillar unwinds all the way down, like a yo-yo. For a solid cylinder the rope's tension is a third of the weight, and the centre falls with acceleration $\frac{2}{3}g$. So $60 = \frac{1}{2}\cdot\frac{2}{3}g\,t^2$, giving $t = \sqrt{3\cdot 60/g} \approx 2.36$ seconds. The only printed answer was signed "C. Bumpkin", which Leybourn's index gives as Landen himself.

41. About 5.37 seconds. While the chain is on the pulley, let x be the distance it has run. The overhang $2 + 2x$ feet drives all 40 feet, so $\ddot{x} = g(2 + 2x)/40$, and from rest $x = \cosh kt - 1$ with $k = \sqrt{g/20}$. The short end reaches the pulley when $x = 19$, so $\cosh kt = 20$, at $t \approx 2.908$ seconds, with speed $k\sinh kt \approx 25.33$ feet a second. The long end is then 40 feet down, and the whole chain falls freely the other 160 feet in a further 2.46 seconds. Allen's printed answer has the same 2.908137 seconds for the first part.

42. Both weights hang 3 feet down, so the middle piece of thread is level and carries no vertical pull. The two outer pieces then carry 3 and 5 ounces of vertical pull, with the same horizontal pull H, so their horizontal spans are $3H/3$ and $3H/5$. Requiring the three pieces to add up to 10 feet gives the 3-ounce weight 1.589 feet in from its tack, 3.395 feet along the thread, and the 5-ounce weight 0.954 feet in from the other, 3.148 feet along. The middle piece is 3.457 feet.

43. About 12.58 feet. For a chain $y = a\cosh(x/a)$ hanging across a span of $2c$, the pull at each tack is the weight per foot times $a\cosh(c/a)$. This is least when $u = c/a$ satisfies $u\tanh u = 1$, so $u \approx 1.1997$. The length is then $2c\sinh u/u \approx 12.58$ feet, the sag is about 3.38 feet, and each tack bears about 0.60 of the chain's weight.

44. About 49.5 feet up, with the rope at 45°. The turning effect about the foot of the tree is the pull times the height h times the sine of the rope's angle, which is $h\sqrt{70^2 - h^2}/70$. This is greatest when $h = 70/\sqrt{2} \approx 49.5$ feet, less than the tree's 60, so it can be done. This was James Ashton's answer.

45. While the rod touches the wall, its centre moves on a circle round the corner, of radius half the rod. Conservation of energy gives the angular speed at angle θ to the floor: $\dot{\theta}^2 = \frac{3g}{l}(\sin\theta_0 - \sin\theta)$, for a rod of length l let go at θ_0. The wall's push equals the mass times the horizontal acceleration of the centre, and this falls to zero when $\sin\theta = \frac{2}{3}\sin\theta_0$. So the top leaves the wall at two-thirds of its starting height. The Diary also asked for the time and where the foot lands. Those depend on the starting angle, which it left general, and the answer by Peter Walton, another of Landen's pupils, is long.

46. Height 56.04 inches, radius 5.426 inches. A solid cone hung by its point swings like a simple pendulum of length $(4h^2 + r^2)/5h$, so it makes $60\sqrt{39.2\cdot 5h/(4h^2 + r^2)}$ swings a minute. Setting this equal to h, with $\frac{1}{3}\pi r^2 h = 1728$, gives T. Todd's printed answer. The question came back in 1795 as Q995, without the volume. Hutton's note says it was then "unlimited": without a second condition there are infinitely many such cones.

47. Let the cylinder have length L and radius r, and hang it at a point of its axis a distance d from its centre of gravity. Its radius of gyration about the centre is given by $k^2 = L^2/12 + r^2/4$. The equivalent simple pendulum has length $d + k^2/d$. This is shortest, and the swing quickest, when $d = k$, and the length is then $2k$. Amicus gave this, with a neat construction by a semicircle.

48. If the ball swings out to an angle α on each side, the tension at angle θ is $W(3\cos\theta - 2\cos\alpha)$, and its sideways part is $W\sin\theta(3\cos\theta - 2\cos\alpha)$. This is greatest where $\cos\theta = \big(\cos\alpha + \sqrt{\cos^2\alpha + 18}\big)/6$, provided that angle is inside the swing; otherwise the greatest pull comes at the end of the swing. For a swing of a quarter circle each way, the greatest pull is $1\frac{1}{2}W$ at 45°. For a swing of 60° each way it is $0.84W$ at about 37°.

49. About 14,460 feet, roughly $2\frac{3}{4}$ miles. The stick, 5 feet long with its end 3 feet out, makes a cone 4 feet high. A stick is not a weight on a string: a uniform rod turning in a cone goes round in the same time as a simple conical pendulum two-thirds as high. So the time of one turn is $2\pi\sqrt{(8/3)/g} \approx 1.809$ seconds, and seven turns is 12.66 seconds of sound. Isaac Saul's printed answer, 14,461.7 feet, uses this two-thirds rule. Taking the stick as a weight on a string would have given 17,700 feet.

50. About 15.3 miles an hour. The coach comes up from behind, so before it passes the sound travels forwards, the same way as both coach and walker. The cracks are then heard at the rate $f(s-w)/(s-u)$, where f is the true rate, u the coach's speed, $w = 4$ the walker's and $s \approx 778.6$ miles an hour that of sound. Afterwards the sound comes back towards the walker, and the rate is $f(s+w)/(s+u)$. Setting the ratio of the two equal to 70/68 and solving for u gives 15.28 miles an hour. The Diary's editor printed answers on two different readings of what the counts meant, and put more in the Supplement.

51. About 2.73 gallons. If x gallons of wine remain, the mixture leaving in a short time dt seconds carries away $x/100$ of a gallon of wine for every gallon, so $dx/dt = -x/1000$. Then $x = 100e^{-t/1000}$, and after an hour $x = 100e^{-3.6} \approx 2.73$ gallons. James Young's printed answer works it with logarithms, as here.

52. By Torricelli's rule the water leaves with speed proportional to $\sqrt{h}$, where h is the depth. So the time to empty is proportional to $\int A(h)\,h^{-1/2}\,dh$, where $A(h)$ is the area of the surface at height h. Put the radius as R. For the cone, $A = \pi R^2(1-h/2R)^2$; for the sphere, $A = \pi(2Rh - h^2)$. Both integrals from 0 to $2R$ come to $\frac{16}{15}\pi R^2\sqrt{2R}$. This is true on any rule in which the speed goes as $\sqrt{h}$, so it holds on the Diary's half-depth rule too.

53. The area of the surface must be proportional to the speed of the outflow, and so to $\sqrt{h}$. The radius therefore goes as $h^{1/4}$: the vessel is a quartic paraboloid, the shape of an ancient water clock. On the Diary's rule, with outflow speed $\sqrt{gh}$, it holds about 91,300 cubic inches, or 323.8 ale gallons (1,496 litres), and is 53.9 inches across the top. Amicus's printed answer is 323.796. With Torricelli's $\sqrt{2gh}$ it holds 457.9 ale gallons. John Dalton of Kendal, later famous for the atomic theory, also sent an answer.

54. About 31 inches. The air already in the gun is at atmospheric pressure. On a 5-inch bore the atmosphere presses with the weight of a column of mercury 30 inches high, about 290 pounds; Christopher Cox took 298. When the sponge stops, the trapped air must push back with $290 + 100$ pounds. By Boyle's law its length is then $120 \times 290/390 \approx 89$ inches, so the sponge goes about 31 inches in. Cox's figures give 30.13 inches.

55. The surface must be one on which every particle pulls the point equally hard along the axis. In polar co-ordinates about the point, that surface is $r^2 = c^2\cos\theta$; in Cartesian form, $(x^2+y^2)^3 = c^4x^2$. It looks like a slightly flattened apple, with the point in the dimple. Its volume is $\frac{4}{15}\pi c^3$, and its widest part is a distance $3^{-3/4}c \approx 0.439c$ from the point. It pulls about 2.6 per cent harder than a sphere of the same mass. This was Amicus's curve.

When the Diary was wrong

Problem 17. The Diary printed the answer to twenty decimal places: 52.063039916251851082. The true value is 52.063039916251850800…, so the last four figures are wrong.

Problem 34. Two solvers disagreed. Merones had 12.2321 leagues, which is right. Mr Powle had 12.2894.

Problem 18. The solvers gave three different lengths: 119.34, 118.67 and 119.5 inches. Only Barlow's first figure is right. The others mis-counted the swings in an hour.

Problem 33. Omicron's printed answer is a circle. It treats the ball as flying straight to the wall and then falling under gravity on the way back. With gravity acting both ways, the locus is the parabola given here.

Problem 15. The Diary gave $60°13′$. We make it about $60°4′$, with the ellipse the question gives.

Problem 42. S. Bamfield's working is right, but the printed middle piece of thread, 3.3951 feet, repeats the figure for one of the outer pieces. The three pieces must add up to 10 feet, so it should be 3.457.

Problem 19. Mr Ash's printed solution used a rule that holds only for one particular angle of the cone, as Leybourn's note says. His number is right, because the true time depends only on the height.

Where the Diary simply worked differently

Problem 32. The half-depth rule for the speed of water leaving a vessel was Newton's first view and appears in Hutton's textbooks. It is not a slip.

Problem 53. The same half-depth rule. Amicus's 323.8 gallons is right on that rule; Torricelli's would give 457.9.

Problem 31. Newton's law of resistance is not the modern one. The question is kept for its method and its setter: it is the only question in volume IV that carries a woman's name.

Problem 13. Our time differs from the Diary's by twelve minutes, and we do not know why.